

DELHI METRO ANALYTICS DASHBOARD

Power BI Project Presentation

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INTRODUCTION

- Analyses Delhi Metro trip and passenger data using Power BI.
- Provides insights into revenue, cost, profit and passenger trends.
- Uses interactive charts, KPI cards and slicers for easy analysis.
- Helps present operational and financial information in a single dashboard.

OBJECTIVES

To analyse passenger trends

To analyse revenue and expenses

To evaluate profitability

To create interactive visualizations

To provide a data-driven overview

Dataset Description

TripID

Unique trip identifier / trip count k

Date

Trip date used for time analysis

From_Station

Origin station

To_Station

Destination station

Passengers

Passenger volume

Fare

Fare value used for average-fare analysis

Revenue

Revenue measure

Total_Cost

Cost measure

Profit

Profit measure

Profit/Loss

Profitability category

Ticket_Type

Passenger/ticket segmentation

Remarks

Operational/status classification

DATA PREPROCESSING

Power Query – Data Preparation

- Handled missing values
- Checked and removed duplicate records
- Corrected data types and formats
- Standardized categorical values
- Cleaned and transformed the raw dataset
- Created custom columns using Power Query

Calculated Columns & Measures

- Created Route using source and destination stations
- Created Revenue-related calculations
- Created Profit/Loss classification
- Created calculated columns for analysis
- Created measures such as Total Revenue, Total Profit and Average Profit per Passenger
- Created profitability measures for dashboard KPIs

DASHBOARD OVERVIEW

DELHI METRO ANALYTICS DASHBOARD

TOTAL PROFIT

127.94M

TOTAL REVENUE

315.39M

PASSENGERS

3M

TOTAL TRIP PERCENT

0.28

AVG PROFIT PER PASSENGER

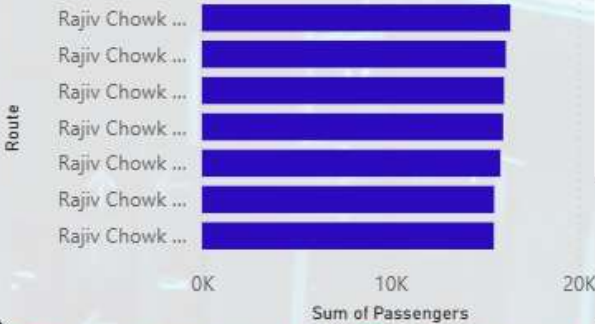
42.64

Profitability & Volume by Ticket Type

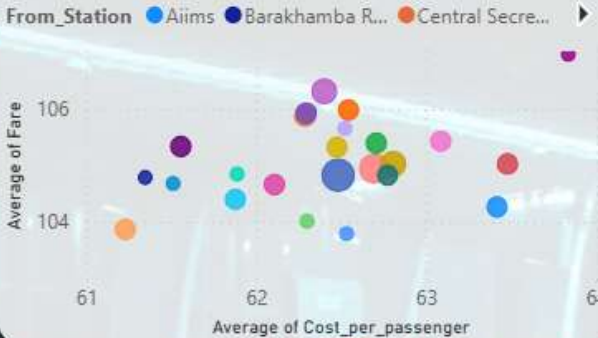
● Total Profit ● Sum of Passengers



Top 10 Busiest Routes



Cost vs. Fare Scatter Analysis



Ticket_Type

- ☐ Return
- ☐ Single
- ☐ Smart Card
- ☐ Tourist Card

Year

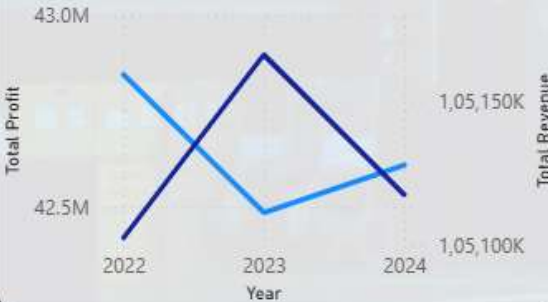
- > ☐ 2022
- > ☐ 2023
- > ☐ 2024

From_Station

- ☐ Aiims
- ☐ Barakhamba ...
- ☐ Central Secre...
- ☐ Chandni Cho...
- ☐ Dilshad Garden

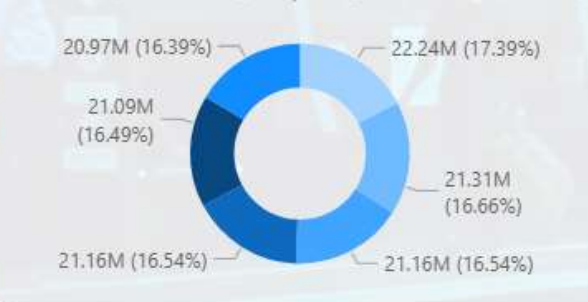
Profit & Revenue Trend

● Total Profit ● Total Revenue



Profit Distribution by Operating Remarks

Remarks ● Unknown ● off-peak ● weekend



From_Station	Return	Single
Shivaji Park	4,86,360.37	5,31,146.60
Rajouri Garden	5,42,961.10	12,70,176.66
Rajiv Chowk	5,38,778.22	37,30,763.44
Punjabi Bagh	5,20,801.93	5,24,888.21
Pragati Maidan	4,85,886.74	5,36,687.89
Old Delhi	5,51,647.20	5,61,997.55
Noida City Centre	4,54,992.53	28,41,926.06
New Delhi	5,48,589.88	21,46,511.34
Total	1,23,16,672.91	3,21,01,678.09

KEY ANALYSIS CRITERIA

Analysis Area	What is Analyzed
Ticket Type Performance	Profit and passenger volume by ticket type
Monthly Performance	Revenue and profit trends over time
Route Demand	Top 10 busiest routes by passengers
Operating Performance	Profit distribution by operating remarks
Cost vs. Fare	Relationship between passenger cost and fare
Station Performance	Profit by station and ticket type

KEY INSIGHTS

- ➡ The number of passengers using Delhi Metro shows an increasing trend. While, the revenue generated decreased.
- ➡ Highest revenue generated from the station Rajiv Chowk. Highest number of passengers are also from Rajiv Chowk. Rajiv Chowk to AIIMS is the most busiest route
- ➡ The number of trips has also increased by time.
- ➡ Kirti Nagar Station has the generated the lowest profit.
- ➡ There is an increase in the profit over years.
- ➡ There are several stations in which revenue generated has increased but there is decrease in profit
- ➡ Every year there is a drop in the metro trip during February- march season
- ➡ Tourist card type of card makes the highest profit.

SUGGESTIONS

-  Increase non-fare revenue
-  Promote Smart Cards
-  Optimize low-demand operations
-  Reduce cost per passenger
-  Improve Tourist Card utilization

CONCLUSION

- ◆ The project provides a comprehensive analysis of Delhi Metro's operational and financial performance.
- ◆ Revenue, cost, profit, passenger volume, trips, and ticket types were analyzed using Power BI.
- ◆ The analysis shows ₹127.58M revenue, ₹75.88M cost, and ₹51.70M profit in the dataset.
- ◆ Passenger and ticket-type analysis provides insights into travel patterns and usage. Station-level analysis helps identify areas with higher passenger and revenue contribution.
- ◆ The dashboard can support revenue optimization, cost control, and better operational planning
- ◆ Overall, the project demonstrates how Power BI can transform raw transportation data into actionable business insights.

REFERENCES

- ➡ Kaggle – *Delhi Metro Dataset – EDA & Data Visualization*
- ➡ Microsoft Learn – *Power BI Documentation*
- ➡ Microsoft Learn – *Power BI Overview*
- ➡ Microsoft Power BI – Official Documentation & Resources

THANK YOU